

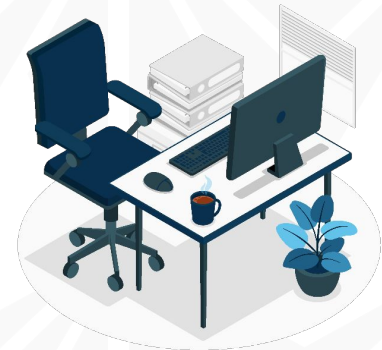


EXPERIMENT TO DETERMINE THE MAGNETIC MOMENT OF A BAR MAGNET



AIM

- To determine the magnetic dipole moment of a bar magnet and its pole strength.





LEARNING OBJECTIVES

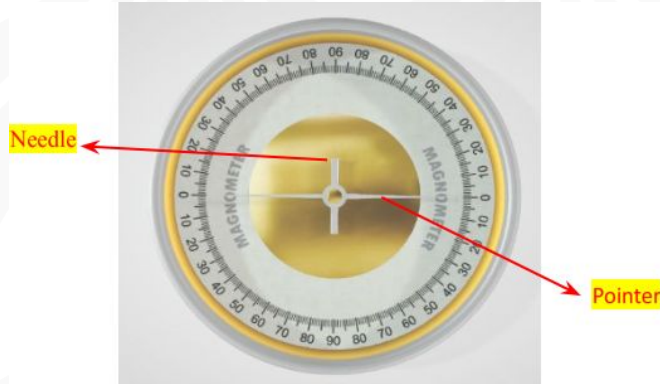
- Explain the construction and operation of the tangent galvanometer.
- Set up an experiment to study the magnetic properties of a bar magnet.
- Determine the value of the pole strength of a bar magnet.





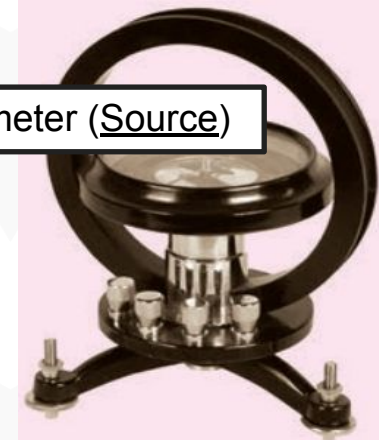
APPARATUS

- Meter stick.
- Tangent galvanometer
- Bar magnet.



Top view of the tangent galvanometer

Tangent galvanometer (Source)





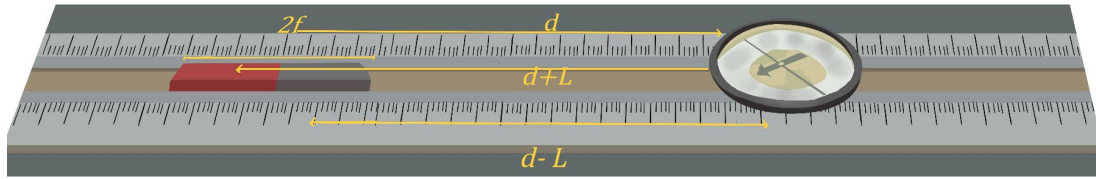
THEORY

- The tangent galvanometer is an early device used to measure electric currents.
- It works based on the principle of the tangent law, which relates the angle of deflection of a magnetic needle in a magnetic field to the strength of an electric current passing through a nearby coil of wire.
- The tangent law relates the tangent of the deflection angle (θ) to the strength of the current (I), $\tan(\theta) \propto I$





THEORY



- In a tangent galvanometer, the magnetic needle is free to rotate.
- It aligns itself with the Earth's magnetic field, pointing north-south due to the horizontal component (H_0) of the Earth's magnetic field.
- This alignment is the basis for measuring electric currents using the galvanometer.



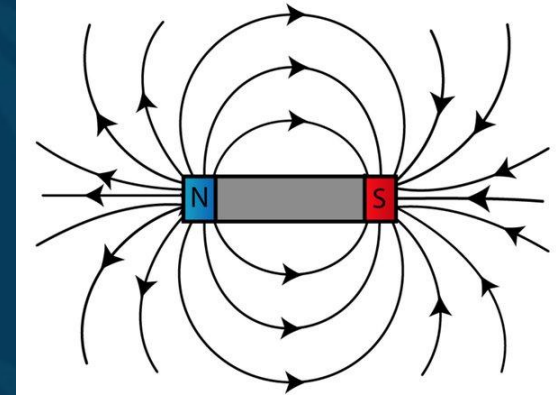


THEORY

- Imagine a bar magnet of length L with its north pole on the left and south pole on the right.
- We want to find the magnetic field H at a point located a distance d to the right of the south pole of the magnet.

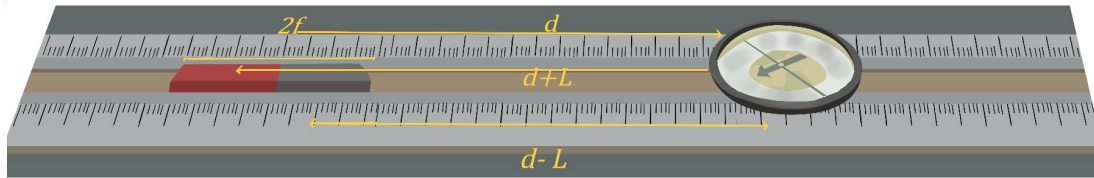
Identify the Magnetic Field Direction

- The magnetic field lines around a bar magnet go from the north pole to the south pole. At the point to the right of the south pole, the magnetic field lines are directed from right to left.





THEORY



H = Magnetic field strength at the point to the right of the magnet.

L = Length of the bar magnet.

d = Distance from the south pole of the magnet to the point where we are measuring the magnetic field.

m = Magnetic pole strength of the magnet.

μ_0 = Permeability of free space ($4\pi \times 10^{-7} \text{ T}\cdot\text{m/A}$).





THEORY

- We can use the Biot-Savart law to calculate the magnetic field at the point.
- The law states:
$$dH = \frac{\mu_0}{4\pi} \cdot \frac{I \cdot dl \cdot \sin(\theta)}{r^2}$$
- *I is the current (which is m for a single magnetic pole),*
- *dl is an element of length along the magnet,*
- *θ is the angle between dl and r (the distance from dl to the point of interest), and*
- *r is the distance from dl to the point*





THEORY

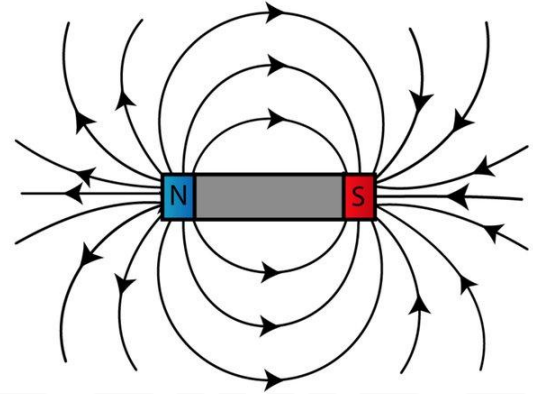
- Since the magnetic field lines are symmetrically arranged around the magnet, we can simplify by considering only the horizontal component of dH because the vertical components will cancel out. hence

$$dH = \frac{\mu_0}{4\pi} \cdot \frac{m \cdot dl}{r^2} \cdot \sin(\theta)$$

Express r in Terms of L and d

For this specific case, r can be expressed as:

$$r = \sqrt{d^2 + (L/2)^2} = \sqrt{d^2 + L^2/4}$$





THEORY

- At the point to the right of the magnet, θ is the angle between dl and r , which is 90 degrees ($\theta=90^\circ$). Therefore, $\sin(\theta)$ is equal to 1.

Integrate Over the Entire Length of the Magnet

- To find the total magnetic field H , we need to integrate the contribution of each element dl along the entire length L of the magnet:

$$H = \int \frac{\mu_0}{4\pi} \cdot \frac{m \cdot dl}{r^2}$$

$$H = \frac{\mu_0}{4\pi} \cdot \int_0^L \frac{m \cdot dl}{(d^2 + L^2/4)}$$

- Simplify and Solve the Integral $H = \frac{\mu_0}{4\pi} \cdot \frac{4mLd}{(d^2 - L^2)^2}$
- We've derived the equation for the magnetic field H at a point to the right of a bar magnet in terms of m , L , d , and μ_0 :





THEORY

$$H = \frac{\mu_0}{4\pi} \cdot \frac{4mLd}{(d^2 - L^2)^2}$$

- This equation represents the strength of the magnetic field at the specified point due to the magnet.

The magnetic dipole moment, $M = \text{Pole strength} \times \text{magnet's length} = 2mL$

In this case, the angle of deflection θ

$$H = H_0 \tan \theta$$

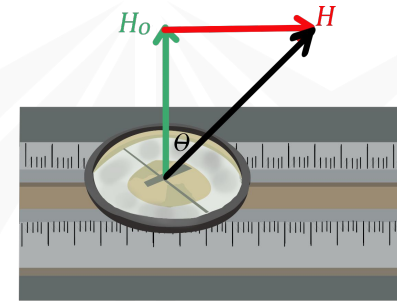
- From (1) and (2), we have

Simplified to:

$$\frac{\mu_0}{4\pi} \frac{2Md}{(d^2 - L^2)^2} = H_0 \tan \theta$$

$$\tan \theta = \left(\frac{M\mu_0}{2\pi H_0} \right) \frac{d}{(d^2 - L^2)^2}$$

$$y = m \quad x$$





MID-LESSON QUESTIONS

Question 1:

What is the principle behind the operation of a tangent galvanometer?

Question 2:

How can you study a bar magnet's magnetic properties?





MID-LESSON ANSWERS

Answer 1: The tangent galvanometer operates based on the principle of the tangent law, which establishes a relationship between the angle of deflection of a magnetic needle within a magnetic field and the strength of an electric current flowing through a nearby coil of wire.

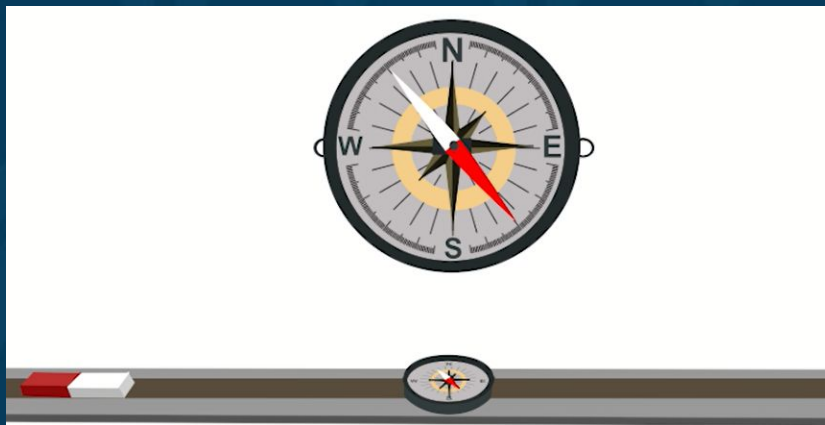
Answer 2: To study a bar magnet's magnetic properties, use a magnetic compass to observe how the magnet affects the direction of the compass needle when placed nearby. The angle of deflection of the compass needle helps us understand the magnet's properties.





PROCEDURE

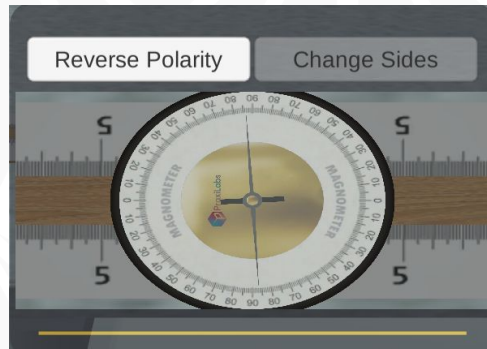
- Place the magnet at the left end of the ruler (it is 50 cm away from the centre of the tangent galvanometer).





PROCEDURE

- Click the Record button to record the deflection angle (θ_1 & θ_2) corresponding to this distance.
- Click the REVERSE POLARITY button.
- Click the Record button to record the deflection angle (θ_3 & θ_4)
- Click the REVERSE POLARITY button to make the magnet restore its original orientation.





PROCEDURE

- Reduce the distance d in step of 5cm (45cm, 40cm, ... 30cm) and repeat the previous steps to record the four deflection angles (θ_1 , θ_2 , θ_3 & θ_4) corresponding to each distance.
- When you finish recording for all of the four angles on the left side for five different distances, the magnet will be moved automatically to the right side.



PROCEDURE

- Repeat all the previous steps to record the corresponding four angles (θ_5 , θ_6 , θ_7 & θ_8) for the same distances you have chosen on the left side.
- The experiment has been completed and an EXCEL sheet will be downloaded containing the data you have recorded.



DATA ANALYSIS

- The downloaded EXCEL sheet will look like this.

Magnetic moment of a bar magnet									
d(cm)		θ_1	θ_2	θ_3	θ_4	θ_5	θ_6	θ_7	θ_8
50		12	12	13	13	12	12	11	11
45		16	16	15	15	15	15	15	15
40		21	21	22	22	21	21	21	21
35		32	32	30	30	32	32	31	31
30		44	44	43	43	45	45	44	44

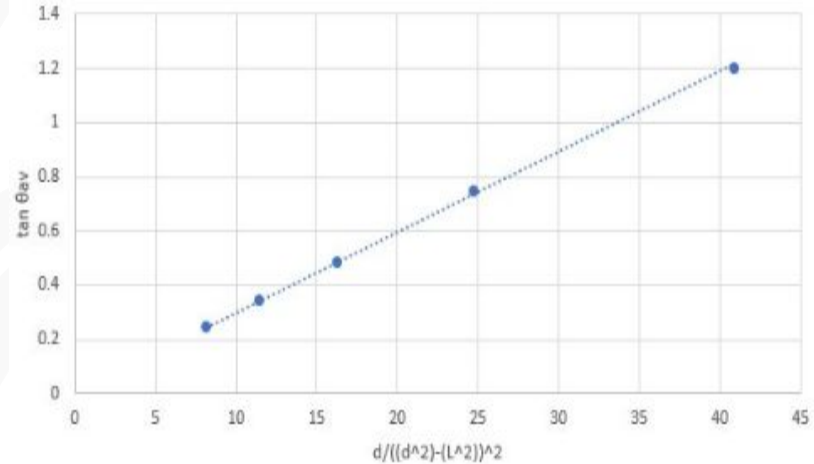


DATA ANALYSIS

- In the next column, calculate the average angle θ_{AV} using

$$\theta_{AV} = (\theta_1 + \theta_2 + \theta_3 + \theta_4 + \theta_5 + \theta_6 + \theta_7 + \theta_8) / 8$$

- Calculate both $\tan \theta_{AV}$ and $[d^2 / (d^2 - L^2)^2]$.
- Plot the graph of $\tan \theta_{AV}$ against $[d^2 / (d^2 - L^2)^2]$ and find its slope.





DATA ANALYSIS

- Calculate the magnetic dipole moment of the bar magnet from:

$$\text{slope} = \frac{M\mu_o}{2\pi H_o}$$

Where $\mu_o = 4\pi \times 10^{-7} \text{ T.m/A}$, and the average value of H_o can be used as $40 \times 10^{-6} \text{ A/m}$

$$M = \frac{2\pi H_o}{\mu_o} \times \text{slope} = 200 \times \text{slope}$$

And the pole strength can be found from:

$$m = \frac{M}{2L} = \frac{M}{0.1}$$



WALK_THROUGH VIDEO





FURTHER READING

- Bueche F. and Hecht E. Schaum's Easy Outlines - College Physics Crash Course. 2000 McGraw-Hill, USA.
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- Sang, D., Hewlett, C., Jones, G., Field, S., & Styles, D. (2020). Cambridge International AS & A Level Physics Practical Workbook. Cambridge University Press.
- Serway, R.A. and Jewett. J.W. (2014). Physics for Scientists and Engineers with Modern Physics, Ninth Edition. Physical Sciences: Mary Finch.





FURTHER READING

- Shukla, R. K., & Srivastava, A. (2006). Practical Physics. New Age International (P) Limited, Publishers.
- Young, H. D. (2012). Sears & Zemansky's college physics. 9th ed. Pearson Education, Inc.
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- <https://openstax.org/books/physics/pages/20-1-magnetic-field-field-lines-and-force>
- <http://www.physicstutorials.org>





THANK
YOU